

Backend API Requirements - EduStream SMS

Technology Stack: PHP Laravel

Target: Flutter App Integration (REST APIs)

Project Overview

The frontend of the School Management System (EduStream) is built in Flutter. We need REST APIs to connect the frontend for live data storage and dashboard analytics. Note: Admin panel features will be discussed in a later phase. The current scope is limited to Students, Invoices, and Dashboard metrics.

1. Dashboard Module

- Endpoint to fetch real-time summary statistics.
- Metrics required: Total Students, Total Revenue (sum of all PAID invoices), and Total Pending Invoices (count of PENDING invoices).

2. Students Module (CRUD)

- GET /students - Fetch all students (needs search query and filter by class/section).
- POST /students - Add a new student. Fields: firstName, lastName, email, rollNumber, grade, section, parentName, phone, parentEmail.
- PUT /students/{id} - Update student details.
- DELETE /students/{id} - Delete a student.

3. Classes & Sections Module

- GET /classes - List all available classes (e.g., Grade 9, Grade 10).
- POST /classes - Create a new class/section dynamically.

4. Invoices & Fees Module

- GET /invoices - Fetch all invoices. Must support filtering by student roll number, billing month, and status (PAID/PENDING).
- POST /invoices/batch - Generate invoices for a specific month for all active students. Logic: Grade 9 = PKR 5500, Grade 10 = PKR 6500, Others = PKR 5000.
- PUT /invoices/{id}/pay - Update invoice status from PENDING to PAID.

API Guidelines for Developer

- All responses should be in JSON format.
- Please include standard pagination for GET requests if possible.
- Use standard HTTP status codes (200 OK, 201 Created, 404 Not Found, etc.).